Inversions

Inverting triads, chords, and intervals rearranges them so that the notes appear in a different order, or pitch. Inversions are essential for successful part-writing, and also for being inventive with the musical materials in a given key.

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❮ 108–109 Analyzing melodies
❮ 126–127 Triads
Cadences 136–137 ❯

What is an inversion?

A melodic inversion turns a melody upside down by raising or lowering notes up or down the octave, so that a given interval becomes its mirror image in relation to an octave. The intervals can be played separately as part of a melody, or together to make an inverted chord.

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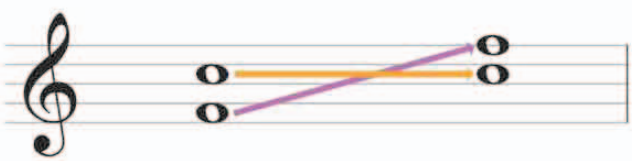

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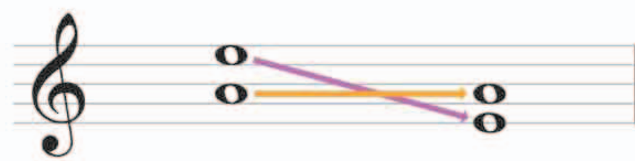


Perfect fifth
The interval between F and C is a perfect fifth.

Perfect fourth
But, when the F is raised an octave, the interval is now a perfect fourth.

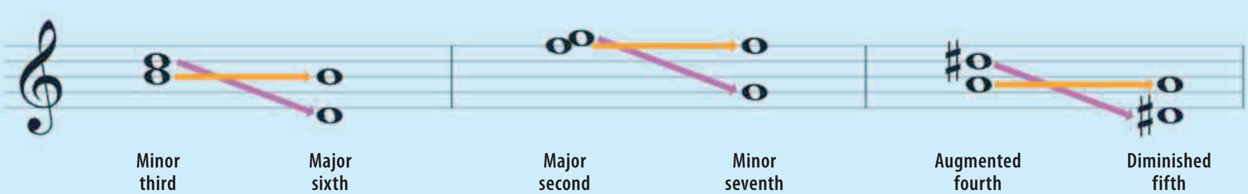
Perfect fifth
Reversing the process, if we take the perfect fifth of A–E and lower the E by an octave.

Perfect fourth
...the E–A interval is a perfect fourth. These are common examples of inversions.

More examples of interval inversions

An octave has 12 half steps. Each original interval when added to its inversion must be equal to 12 half steps.

◁ **Intervals inverted upward**
The inversion of a major third (four half steps) is a minor sixth (eight half steps). The interval of an augmented fourth—tritone—or a diminished fifth (depending on the harmonic context), has six half steps. So its inversion is also six half steps apart.



△ **Intervals inverted downward**
The inversion of a minor third (three half steps) is a major sixth (nine half steps). A major second (F–G here, two half steps apart) becomes a minor seventh (ten half steps).